

Swasthya Avishkar Hackathon – 2025

Adverse Drug Interaction Prediction and Clinical Explanation using Graph Neural Networks and Medical LLMs

Track: AI Pipelines for Drug Repurposing, Efficacy Prediction, and Development

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1. Problem Statement

Adverse Drug Interactions (DDIs) are a major cause of medical complications, clinical emergencies, and avoidable hospitalizations.

Existing systems rely heavily on static rule-based checks, outdated drug databases, or manually curated toxicity lists. These approaches:

- Fail to detect novel or previously unseen DDIs
- Lack explainability, leaving clinicians unaware of why a pair is risky
- Provide no way to explore risks based on symptoms, clinical effects, or mechanisms

Healthcare providers need an intelligent, scalable system that can predict harmful drug–drug interactions proactively, and explain the underlying biological reasoning in a clinically meaningful way.

2. Technology Used in Our Solution

Graph Neural Networks (GNN – PyTorch Geometric): Learns complex biochemical and molecular patterns between drug components to predict harmful interactions.

Graph Construction Pipeline: Converts drug structures into a toxin-component graph, enabling structural-level reasoning.

Vector Database (ChromaDB): Stores 200 high-risk GNN-predicted interactions with embeddings for semantic search.

LLM Layer (Local LLM): Generates clinical, human-readable explanations for each predicted interaction.

Interactive Dashboard (Streamlit + PyVis): Allows doctors or analysts to search interactions, explore data, and visualize the drug interaction graph.

3. Overview of the Solution Approach

Step 1 — Graph Construction

We processed raw drug data by decomposing each drug into molecular substructures (toxins). Each toxin becomes a node, and relations form edges, creating a drug–toxin interaction graph. This allows the system to detect risk even when two drugs don't appear related at a whole-drug level.

Step 2 — GNN Risk Prediction

We trained a GCN-based link prediction model that learns dangerous component interactions, uncovers toxic patterns, and identifies hidden drug–drug risks beyond rule-based systems. The model predicts a risk score (0–1) for any pair of drugs.

Step 3 — LLM-Based Clinical Explanation

The top 200 high-risk interactions are LLM-analyzed to generate mechanism insights, symptom risks, and clinical reasoning for real-world usability

Step 4 — Searchable Vector Database

ChromaDB enables fast semantic search across drugs, symptoms, mechanisms, and explanations.

Step 5 — Interactive Dashboard (Final Product Layer)

An interactive dashboard lets clinicians search by symptoms or drug names, browse high-risk interactions, and visualize drug networks through PyVis.